Self-Reflection: Project BattleBot

Reflected by

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Introduction

As a beginner with limited prior knowledge in robotic programming, I still remember the first day I got the BB008 robot kit. At that time, I didn't even know which cable to use to connect the robot to my laptop for programming. After getting a clue about the programming cable from the teacher, my teammate Michale and I spent two more hours that day trying to figure out how to wire the jumper cable and program the BB008 to blink for LED testing, although we read the manual several times. Again, we got this clue from our classmates and learned that we had to read and study the manual carefully. Since the first day, I have felt a lot of enjoyment and learned new things a lot in this course, and below is my self-reflection.

Situation

The project Battlebot required both hardware and software integration to complete weekly assignments and a fully functional autonomous robot for race day. With the limited background knowledge about robotic programing and hardware systems mentioned above, this project was exciting and challenging at the same time for me.

Task

In the project, BB008 was the name of my team's robot, which needed to setup hardware setting programmed by using C++ in the Arduino IDE to complete

weekly assignments, including blinking LEDs, traffic lights, basic movements, object avoidance, line following, and object grabbing. Finally, all these tasks were combined into a complete game logic sequence to enable our BB008 to perform smoothly on a curve line-following track during race day.

Action

After we realized that we needed to study the manual step by step, our progress became easier, starting with the first assignment, which was the blinking system. For each assignment, we studied the manual carefully and followed the teacher's instructions, workshops, and feedback. In our working process, my sub-teammate (Michael) and I always keep communication and collaboration to make our BB008 achieve each weekly assignment and prepare it ready for the race day. So, we always worked, learned, designed the logic, tested the robot, discussed, and solved problems together. We regularly shared our problems with teachers, teammates, and classmates. We also exchanged our solutions and discussed how we programmed and developed our logic with them as well. In addition, we used AI tools such as ChatGPT to support our learning and help us solve problems. However, sometimes ChatGPT created more problems. In addition, I took the robot home every week to continue working and learn more both hardware and programming. I even copied the starting point of the real racetrack and built it at home for my own practice. However, I found out that it did not always perform the same in real conditions due to friction on different materials and surfaces. Therefore, calibration was always needed.

Result

As a result, I felt proud when I looked back at the first day of this course, when I stood lost in a forest of jumper wires and circuits. Our BB008 successfully

completed all the required tasks, including the weekly assignments and the full game logic for the curve line-following track before the test day and the race day. The way we worked consistently on the project as a strong team bond finally paid off. I found It was always challenging and exciting to test and see how BB008 performed and sometimes felt annoyed to see undesired performance or new problems occurred after being able to solve previous ones. Patience and resilience became new skills I gained coincidently from this project. Overall, I really enjoyed the learning process in this course. It helped me improve my logical thinking, as well as my skills in hardware and logic design, experimenting, testing, and problem-solving.

Reflection

I found that this project was not only about technical coding, but also about logical and systematic thinking, experimentation, and problem-solving. Moreover, I developed professional skills such as teamwork, communication, collaboration, patience, and resilience. I always called my BB008 "a naughty boy" because he did not behave the way we expected (finally and luckily, he became a good boy on the test day), but because of a naughty boy, I have learned a lot from him. It also taught me the importance of testing to ensure the robot performs accurately. The more problems I faced, the more learning and experience I gained. I became more logical, systematic in solving problems, patient and resilient. I would also like to express my gratitude to the teachers for their guidance and feedback, to my teammate Michael O. Baoteng, and to all IT-1D team members and classmates for their support and knowledge sharing.

Transfer

In future projects, I would like to continue improving my robotic programming skills such as C++. Although I can design the logic to make the robot operate in a complete sequence, I still am not yet able to write the full code by myself. I still rely on AI tools to help me in this process. In addition, I learned that it would be better to use version control, such as GitHub, to manage, store and track the code versions properly. This way will help avoid confusion of file version and file loss. Finally, if I have opportunity in the future, I would like to conduct my own project with robots. I found that this course inspired and motivated me to continue experimenting with robotics.